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Building Grade ACP sheets

INSTALLATION & MAINTENANCE INSTRUCTIONS

FEITENG®

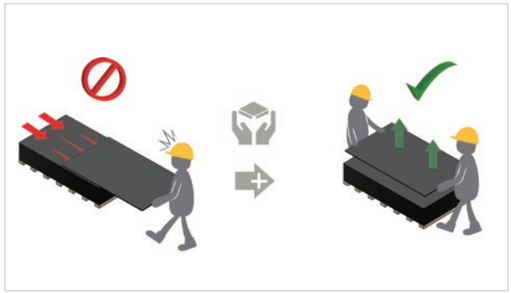
DIPPER

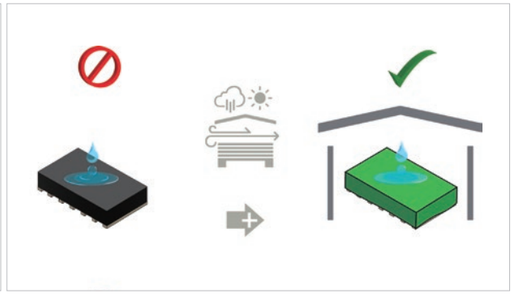


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FEITENG® DIPPER SERIES

Storage & Handling









Film Protect

The protective film on FEITENG® metal composite panels plays an essential role during processing, transportation and installation. They prevent the metal surface from accumulating dust and stains. Long time exposure to sunlight can cause the protective film to lose its adhesive properties and become difficult to remove. Therefore, it is crucial to remove the protective film within 45 days after the panels are installed on the wall. Additionally, when installing FEITENG® metal composite panels, ensure that the arrow markings on the protective film are aligned in the same direction and pointing upward.



FEITENG® DIPPER SERIES

Cleaning & Maintenance

1. The surface of FEITENG® metal composite panels requires regular professional cleaning and maintenance. Such cleaning is not only to keep the building looking clean and attractive but also to regularly remove harmful dirt accumulated on the surface, ensuring the long-term quality of the panels.

2. The frequency of cleaning largely depends on the environmental conditions of the project location and the actual level of dirtiness on the panel surface. We recommend cleaning the exterior walls of buildings at least once a year, while interior walls should be cleaned based on the actual situation of the panel cleanness.

3. The cleaning of curtain walls should be carried out manually or with appropriate cleaning equipment, moving from top to bottom. Avoid using any abrasive materials to scrub the surface. It is recommended to use soft cloths, sponges, or brushes made of non-abrasive materials, and it is best to clean in a consistent direction during the process.

4. Do not clean panel surfaces when their temperature is above 40°C, as rapid water evaporation can affect the metal finish. Always use an appropriate metal cleaning agent and ensure the detergent is neutral. Avoid using strong alkaline detergents (e.g., potassium hydroxide, sodium hydroxide, or sodium carbonate), strong acidic detergents (e.g., nitric acid or sulfuric acid), abrasive cleaners, or cleaners that are corrosive to painted surfaces (e.g., polishing paste, acetone, methyl ethyl ketone, or thinner). Proper cleaning practices help maintain the panel's finish and durability.

Examples of commonly used detergents:

Product name
Turco 5948DPM THICK

Manufacturer
Henkel Surface Technologies

Product name
Tegee, TG22

Manufacturer
Tegee Chemie Bremen Gmbh

Product name
Cream Cleaner K2007

Manufacturer
Klenco (S) Pte Ltd



5. The specific cleaning steps are as follows:
   1) Rinse the panel surface thoroughly with plenty of water.
   2) Use a soft brush soaked in diluted detergent to gently scrub the panel surface.
   3) Rinse the surface again with plenty of water to remove dirt.
   4) Inspect the surface and use detergent to focus on areas that are not fully cleaned.
   5) Rinse the surface with clean water until all detergent is completely washed off.

6. Before starting large-scale cleaning, it is recommended to test a small section of the panel to ensure the method is safe and effective.

7. During maintenance, avoid using hard tools for scraping, hammering or striking, and prevent impacts with hard objects to avoid damage on the panel surface.

FEITENG® DIPPER SERIES

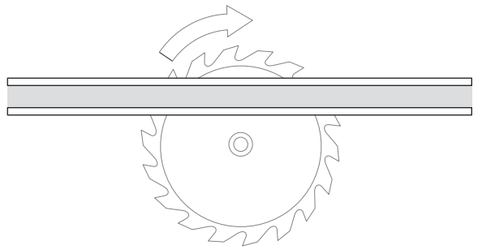
Processing Method

FEITENG® metal composite panels can be easily processed using standard metalworking tools for cutting, punching, grooving and bending. Based on the desired architectural façade design, they can achieve various shapes, including diverse curved surface designs.

Cutting

FEITENG® panels can be cut using common tools such as CNC machines, handheld cutting tools and shearing machines, each suited for different applications: shearing machines are ideal for large-scale production but are limited to simple rectangular cuts, handheld cutting tools are perfect for small-scale processing and on-site modifications, and CNC machines can cut various shapes, including intricate hollow patterns (with hollow sections not exceeding 40% of the panel surface), offering high precision and engraving capabilities.

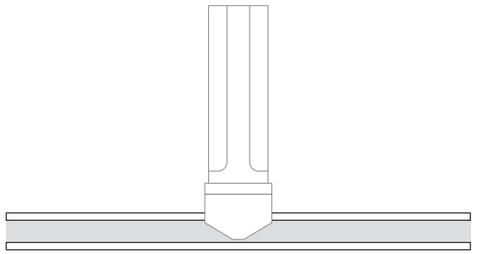
Note: It is important to select the appropriate cutting technique and equipment based on specific requirements, while also adhering to proper operational procedures and regularly maintaining the cutting equipment to ensure optimal performance and cutting quality.



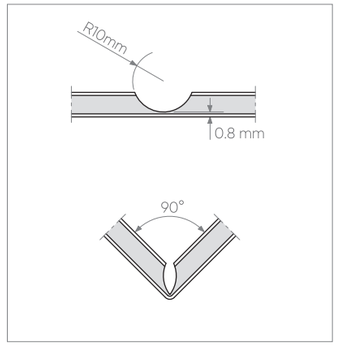
Milling

FEITENG® panels can be milled using CNC machines, BOSCH® trim routers, and BOSCH® handheld circular saws. By changing the corresponding milling tools, different shapes of grooves can be milled into the back of the panel (as shown below). The milling depth should be determined based on the bending requirements.

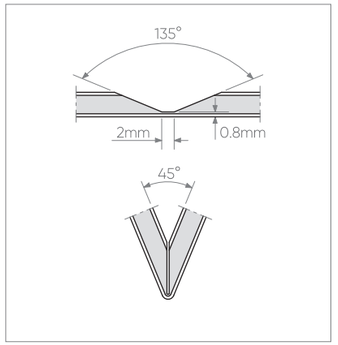
Note: It is generally recommended to leave at least 0.3mm of core material thickness. The middle core layer should not be entirely milled away, as this would reduce the robustness of the metal surface after bending, potentially shortening the lifespan of the panel.



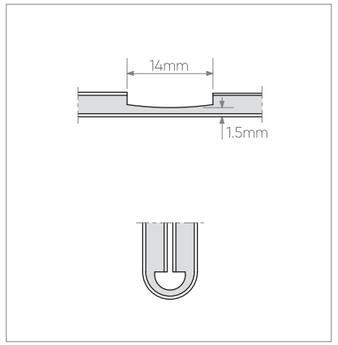
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Spherical milling slot for 90° bends.



V-shaped milling slot for 135° bends.



Arc milling slot for full 180° bends (not possible on FEITENG® A2 panel).

Recommended Milling Parameters

Composite Panel	Operation	Rotation Speed (R.P.M.)	Displacement Speed (mm/min)
FEITENG® DIPPER SERIES A2	Cutting	22.000-24.000	8.000-10.000
	Milling	24.000	10.000-12.000
FEITENG® DIPPER SERIES FR	Cutting	24.000	12.000-14.000
	Milling	24.000	15.000-18.000

Note: these values are indicative only and are intended for use with FEITENG® recommended cutters. For use with other tools or depending on the type of machine, it is necessary to adjust them.

Recommended tools core A2:

Cutting Tool: Cutter Ø6 mm with diamond inserts (STB-FRESA6PCD)

Milling Tool: Spherical milling machine for fine diamond inserts (STB-FRESA10PCD)

Recommended tools core FR:

Cutting Tool: Cutter Ø6 mm with diamond inserts (STB-FD6X80X3.8)

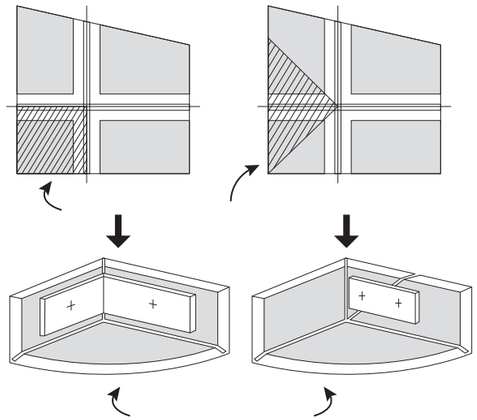
Milling Tool: Spherical milling cutter for fine diamond coating (STB-FEMD-10x72)

Processing Method

Bending

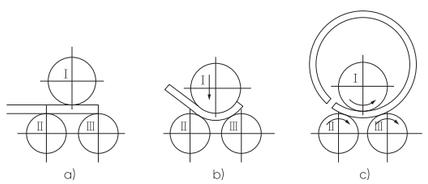
Metal composite panels cannot be directly bent; before bending, a groove must be milled at the bending location. After milling, clean the debris from the groove. When manually bending, use tools to bend the edge in one go, avoiding repeated bending to prevent damage to the surface material. Prior to bending, cut notches around the edges of the panel. After bending, reinforce the areas between the bends using aluminum fittings and rivets, as shown in the diagram below.

Note: The area where the bending tool contacts the metal composite panel should be protected with flexible material. During the bending process, the force should be applied evenly to avoid deformation of the panel.



Coiling Circle

The rolling of metal composite panels requires specialized metal bending equipment, such as bending machines, three-axis bending machines and other bending devices. Before processing, a trial run is needed to determine the number of rolls and the rebound coefficient. During processing, the decorative surface is protected using flexible material between the rollers. Below is the working principle of the symmetric three-roll bending machine:



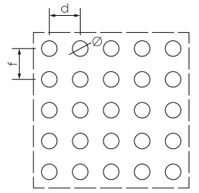
The suitable bending radius for FEITENG® DIPPER SERIES metal composite panel needs to be derived from the geometric characteristics of the arc and the mechanical principles during the bending process. The following are several common design scenarios:

For simple circular arc bending, the following formula can be used to calculate the bending radius:
R = (h^2 + (L/2)^2) / (2h)
Among them, R represents the bending radius, L represents the chord length of the arc, and h represents the height of the arc.
1. This formula is suitable for situations where the ratio between the length and height of the arc is moderate. When the length of the arc is much greater than the height, it can be approximated as R ≈ L2/8h.
2. For the relationship between the bending radius of an arc and the elastic modulus, bending force, and section moment of inertia of the material, the following formula can be used to calculate:
R = (E×I)/M.
Among them, R represents the bending radius, E represents the elastic modulus of the material, I represents the cross-sectional moment of inertia, and M represents the bending force.

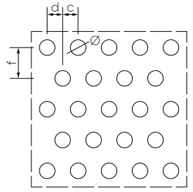
Name	Parameter	Notes
Class A FR	R≥1200mm	Requires heating during processing
FR	R≥380mm	

Stamping

The surface of FEITENG® metal composite panels can be stamped using specialized CNC machines. The shapes of the punched holes can include round holes, square holes, elongated round holes, hexagonal holes, fish-scale holes, etc. Common arrangements include straight alignment and staggered alignment. Detailed stamping parameters are shown in the diagram below.



Distribute Even



Distribute Cross

Classification	Parameter	Notes
Hole diameter (a)	≥4.5mm	
Hole spacing (d, f, e)	≥10mm	
Transmittance	≤35%	Transmittance=(punching area/total area) x 100%

Riveting

The connection of FEITENG® metal composite panels is done using rivets. Commonly used rivet specifications include M411, M511, although other models can be chosen based on requirements. Rivet connections require holes to be drilled in the composite panel. The distance from the center of the hole to the edge of the panel should not be less than twice the panel thickness. A gap of no more than 0.3mm should be left between the hole and the rivet to avoid compressing and deforming the composite panel.

Adhesion

When the size of the FEITENG® metal composite panel is relatively large, structural reinforcement should be applied to the back of the panel based on the design wind load value of the project. The reinforcement material should be aluminum profiles (which can be U-shaped, C-shaped or square tubes). The aluminum profiles and the back of the panel should be effectively bonded using a two-component silicone structural adhesive (and, if necessary, combined with 3M double-sided adhesive).

Suggest using the following adhesives:

DOW CORNING-DC995
(single component structural adhesive)

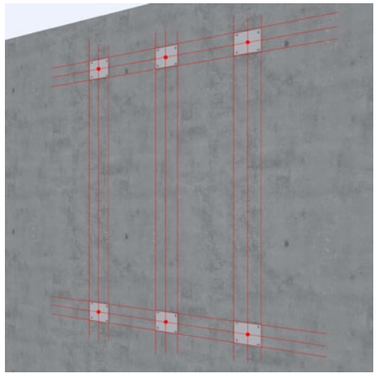
DOW CORNING-DC791
(silicone weather resistant adhesive)

DOW CORNING-DC993N
(two component structural adhesive)

SIKA-SG20
(single component structural adhesive)

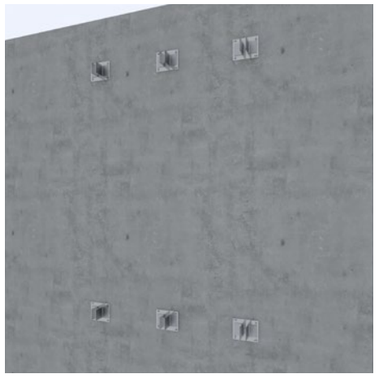
SIKA-WS305CN
(silicone weather resistant adhesive)

Installation Steps



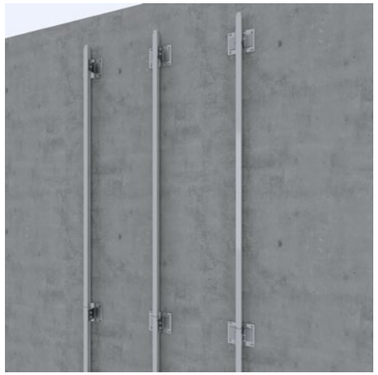
1. Determine the position and size

Based on the curtain wall, construction drawings, architectural drawings, structural drawings, and detailed grid drawings, combined with the elevation control points provided by the civil engineering unit, accurately set out the entry and exit lines and axis positions;



2. Install the embedded parts

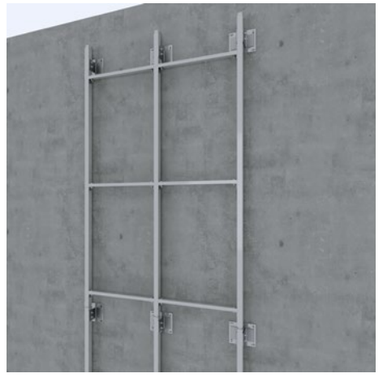
Install steel connectors based on the centerline and grid lines marked during the laying out process. The installation sequence is to first spot weld and fix them, check and verify that they meet the requirements, and then fully weld and fix the steel connectors. The height and left and right of the steel connectors should be controlled within 2-3mm;



3. Column installation

The installation of the column should be between the two adapters of the fixed support, and the column should be fixed with stainless steel tension bolts according to the installation elevation requirements. The deviation of the column axis in the front and back and left and right directions should not exceed 2mm and 3mm, respectively;

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4. Installation of crossbeam

Install the connectors and elastic rubber pads at both ends of the crossbeam at the predetermined position of the column. The installation should be firm and the joints should be tight. The horizontal elevation deviation between adjacent crossbeams should not exceed 1mm. The elevation deviation on the same floor: when the width of a curtain wall is less than or equal to 35m, it should not exceed 5mm; When the width of a curtain wall is greater than 35m, it should not exceed 7mm;

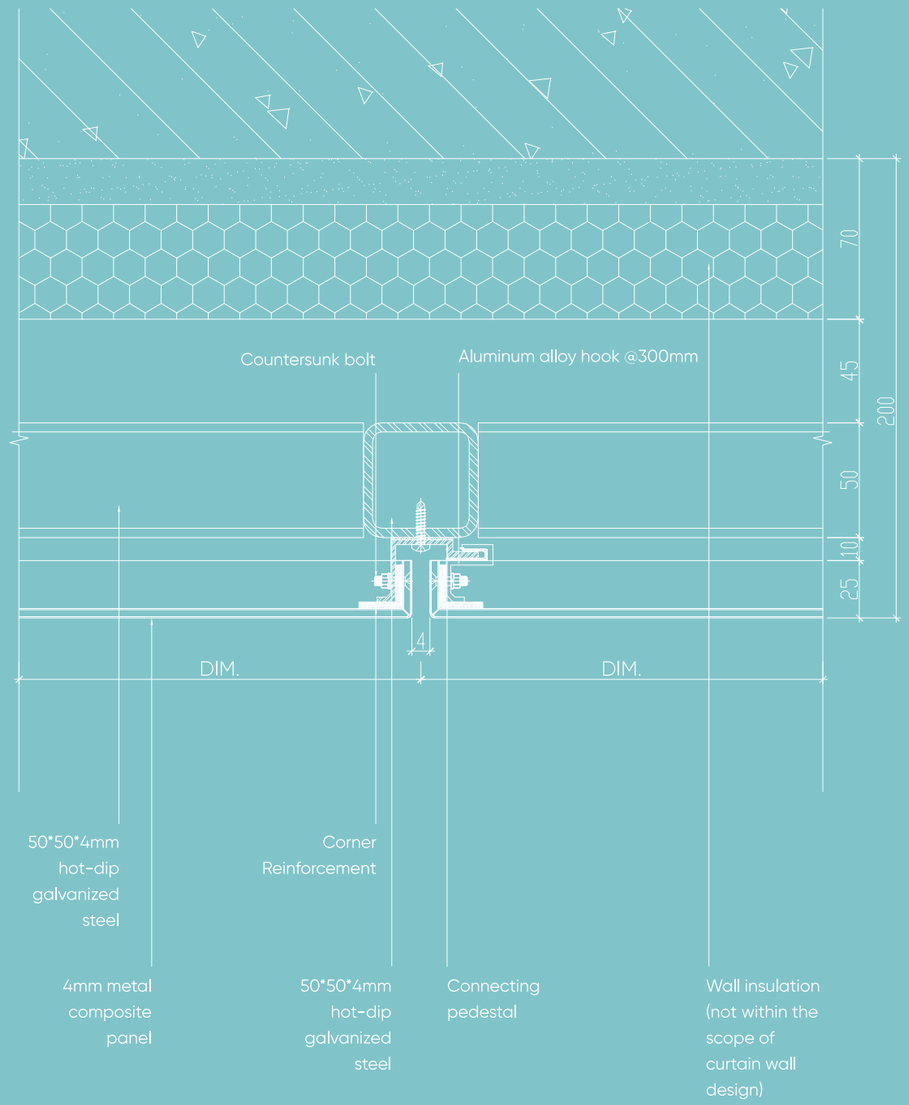


5. Panel installation

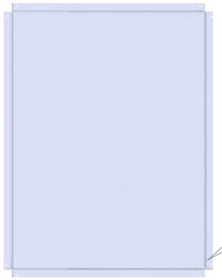
Install the metal composite panel layer by layer onto the frame according to the design requirements. During the installation process, attention should be paid to the flatness and verticality of the metal panel, as well as the connection method and sealing treatment with the frame.

Open Curtain Wall

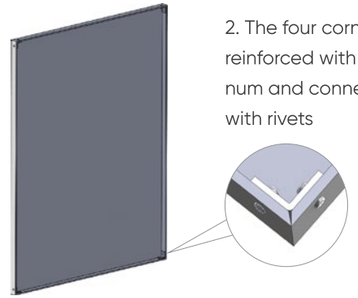
Hanging Code Installation Node



FEITENG

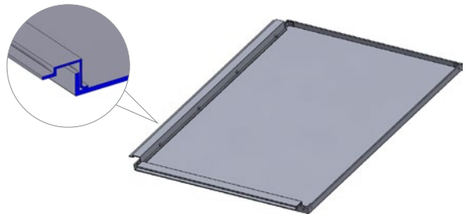


1. Cutting, slotting, bending, forming and placing on the work-bench.

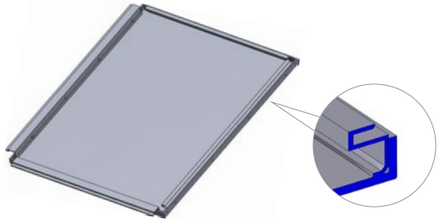


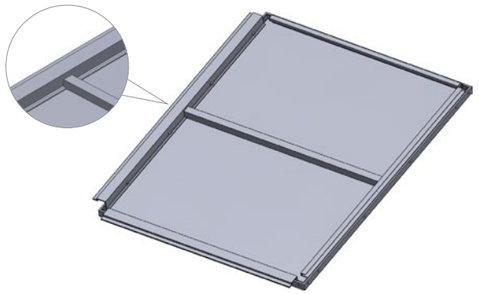
2. The four corners are reinforced with alumi-num and connected with rivets

3. One side of the long folded edge is riveted to the sub profile



4. The other side of the long folded edge is riveted to the mother profile





5. Use structural adhesive to stick the reinforcement ribs to the surrounding profiles



6. Insert one end of the composite panel sub profile into the adjacent composite panel master profile



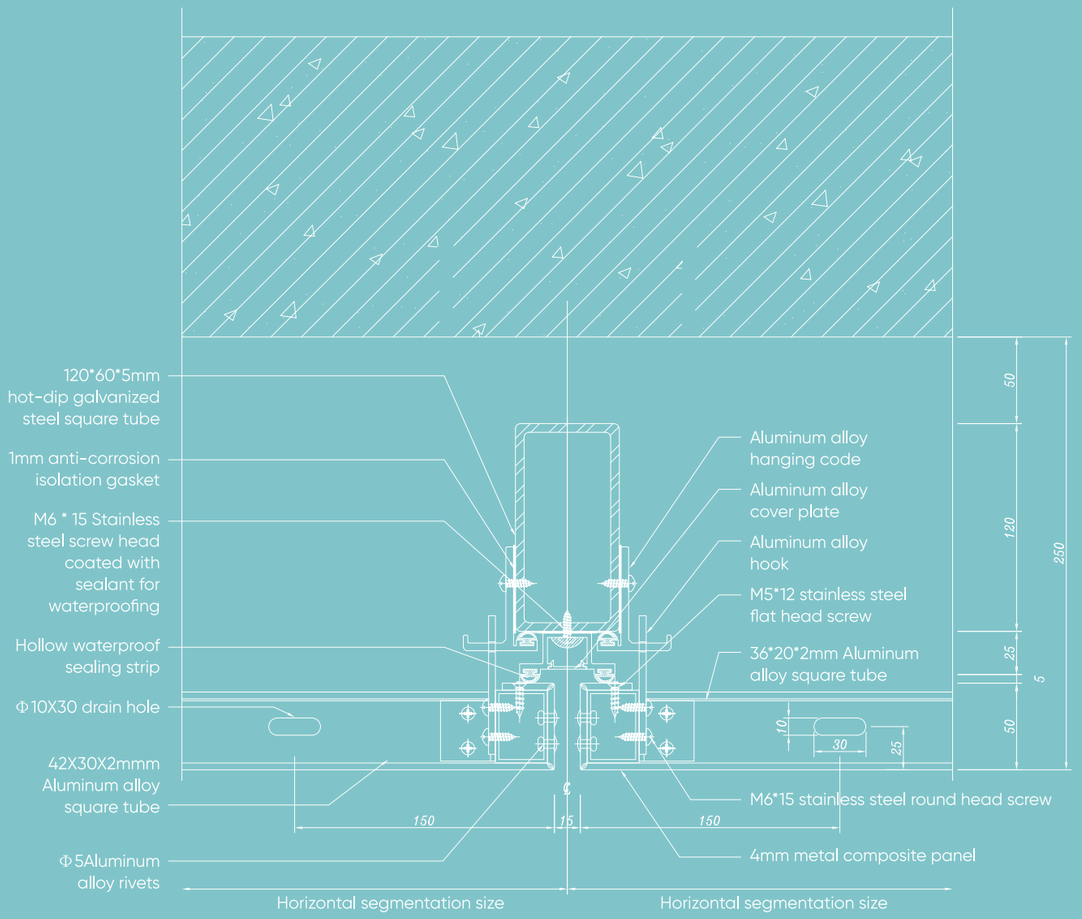
7. After adjusting the position, fix the master profile on the main keel with self tapping screws

Precautions

1. If M5 rivets are used for riveting holes, it is recommended to open 5.2-5.5 holes;
2. When the panel width is less than 600mm, no reinforcing bars shall be arranged. If the width of the panel exceeds 600mm, the spacing between the reinforcing bars shall be arranged at 500mm, and for panels wider than 1500mm, it shall be determined separately. The reinforcement bars are generally arranged perpendicular to the long side.

Open Curtain Wall

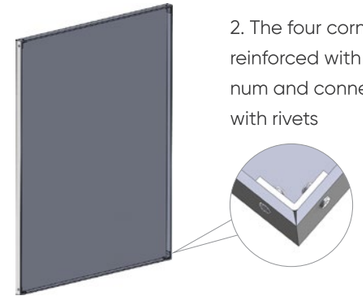
Sub Frame Installation Node



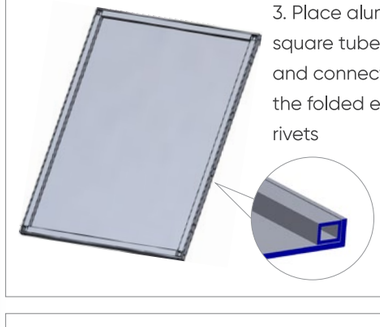
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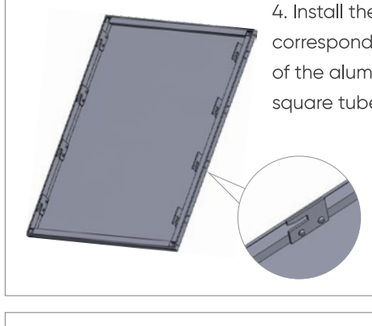
1. Cutting, slotting, bending, forming and placing on the work-bench.



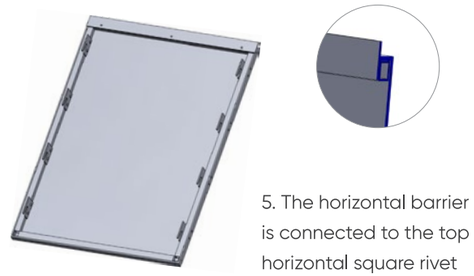
2. The four corners are reinforced with alumi-num and connected with rivets



3. Place aluminum alloy square tube profiles and connect them to the folded edges with rivets



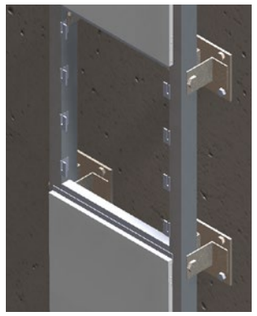
4. Install the hook at the corresponding position of the aluminum alloy square tube profile



5. The horizontal barrier is connected to the top horizontal square rivet



6. Use structural adhesive to stick the reinforcement ribs to the surrounding profiles



7. Install aluminum alloy hanging codes on the vertical keel using self tapping screws



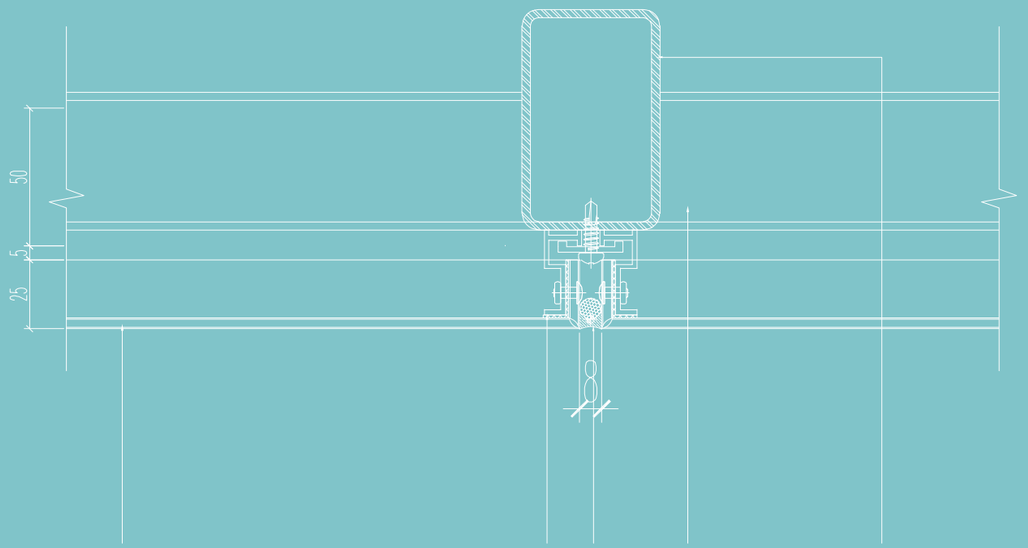
8. Install vertical gap waterproof aluminum alloy cover plate with self tapping screws

Precautions

1. If M5 rivets are used for riveting holes, it is recommended to open 5.2-5.5 holes;
2. When the panel width is less than 600mm, no reinforcing bars shall be arranged. If the width of the panel exceeds 600mm, the spacing between the reinforcing bars shall be arranged at 500mm, and for panels wider than 1500mm, it shall be determined separately. The reinforcement bars are generally arranged perpendicular to the long side.

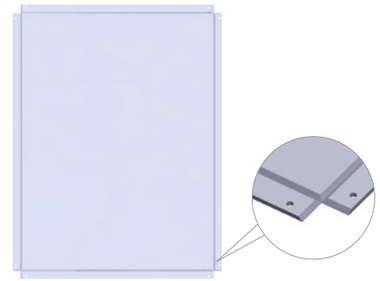
Closed Curtain Wall

Sub Frame Installation Node

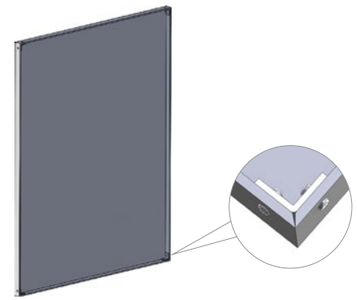


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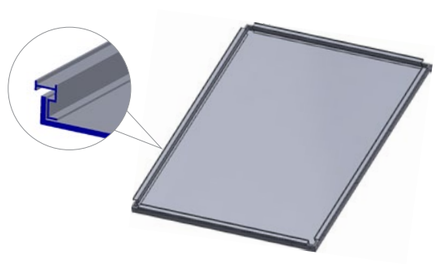
1. Cutting, slotting, bending, forming and placing on the workbench



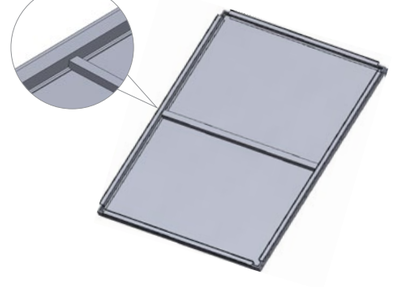
2. The four corners are reinforced with aluminum and connected with rivets



3. Place aluminum alloy special profiles and connect them with rivets and folded edges



4. Use structural adhesive to stick the reinforcement ribs to the surrounding profiles





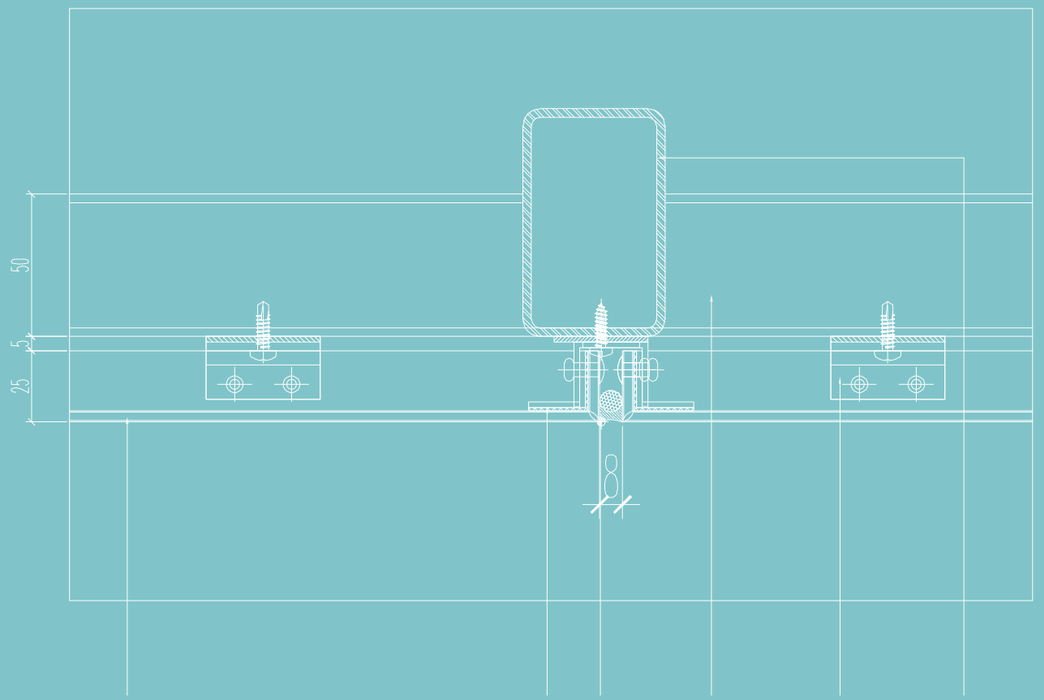
5. Insert the block shaped aluminum alloy pressure block through the gap, fix the pressure block with self tapping screws to make the panel profile tightly adhere to the steel keel and leave the design gap



6. Fill the gap with a round foam rod, paste masking paper on the panel, apply sealant, and then tear off the masking paper

Closed Curtain Wall

Plug in Installation Node



4mm metal composite panel

2mm aluminum subframe structural adhesive bonding

Silicone sealant

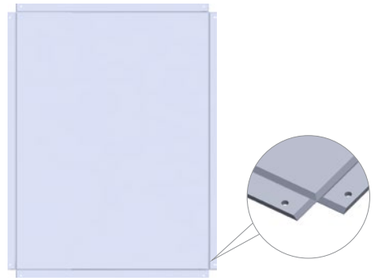
Steel crossbeam square tube

Aluminum alloy corner code

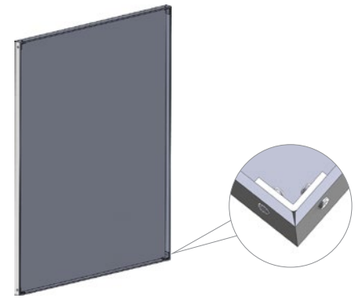
Steel column square tube

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1. Cutting, slotting, bending, forming and placing on the workbench



2. The four corners are reinforced with aluminum and connected with rivets



3. Place aluminum alloy corner aluminum reinforcement components and use rivets to connect and install corner brackets



4. Use structural adhesive to stick the reinforcement ribs to the surrounding profiles





5. After adjusting the horizontal and vertical positions of the panel, use self tapping screws to fix the corner brackets and leave design gaps.



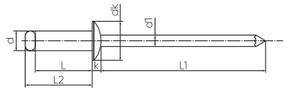
6. Fill the gap with a round foam rod, paste masking paper on the panel, apply sealant, and then tear off the masking paper

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Accessories

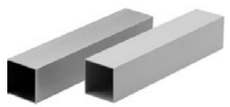
Pull Rivet

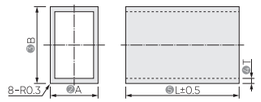




	Specification	3mm	3.2mm	4mm	5mm	6mm
d	Min	2.93	3.11	3.91	4.91	5.91
	Max	3.07	3.29	4.09	5.09	6.09
dk	Min	5.76	5.76	7.71	9.31	11.65
	Max	6.24	6.24	8.29	9.89	12.35
k	Max	1.4	1.4	1.7	2	2.5
d1	≈	1.8	1.8	2.18	2.8	3.6
L2	Max	L+3.4	L+3.5	L+3.5	L+4.0	L+4.5
L1	Min	26	26	27	27	31
L		7-19	7-19	8-20	9-34	10-48

Aluminum Square Pipe

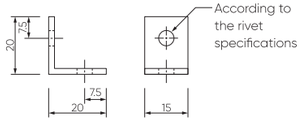




Specification			②T	③L Designated unit 1mm	Weight kg/m	Cross area mm²	Performance	
①Type	②A	③B					ℓx×10⁴	ℓy×10⁴
HFHQ	15	15	1.5	300-2000	0.220	81	0.249	0.249
		30			0.342	126	0.454	1.406
	20	20	2		0.390	144	0.787	0.787
		40			0.607	224	1.437	4.445
	25	25			0.497	184	1.634	1.634
		50			0.766	284	2.96	9.007
	30	30			0.607	224	2.941	2.941
		60			0.934	344	5.297	15.94
	40	40			0.824	304	7.336	7.336
		80			1.648	464	13.11	38.97
	50	50			1.036	384	14.77	14.77

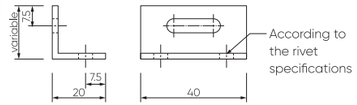
Aluminum Connector



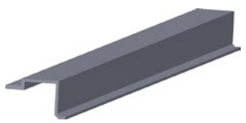


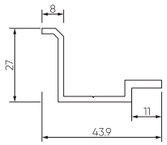
Mounting Angle Code



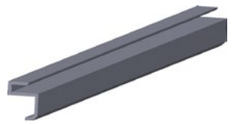


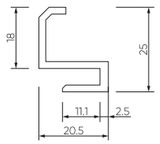
FEITENG Sub Frame 1





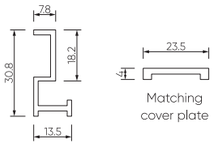
FEITENG Sub Frame 2





FEITENG Sub Frame 3





Hanging Corner Code



